

Homework Assignment 3: Critical Systems by Diffusion, P_N and S_N

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Question 1: Critical radius of a reflected sphere

Question 1 is the one *diffusion* question of this assignment — the S_N machinery set out before Question 3 belongs to Questions 3–5 — and the only one with two materials. A multiplying core of radius R_C is wrapped in $d \in \{1, 2, 3, 10\}$ mean free paths of reflector, and the cross sections are again the Question 5 table of Assignment 1.

The two regions

Every radius here is measured in mean free paths of the medium it sits in:

$$a = \Sigma_{t,C} R_C, \quad b = \Sigma_{t,R} R_R = a + d \quad (1)$$

the dimensionless core radius and the dimensionless outer radius of the reflector, whose width is the given d .

In both regions the substitution $u(r) = r\phi(r)$, which uses $\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = \frac{1}{r} \frac{d^2 u}{dr^2}$, turns the spherical diffusion equation into a plane one:

$$\frac{d^2 u}{dr^2} - \frac{(1-c)\Sigma_t^2}{D_0} u = 0 \quad (2)$$

The core has $c_C > 1$, so the coefficient is negative and the solution oscillates; finiteness at the origin selects the sine. The reflector has $c_R \leq 1$ and decays, vanishing $z_0(c_R)$ mean free paths past b . The relaxation rate is $k_0(c_C)$ per mean free path in the core and $1/\nu_0(c_R)$ in the reflector, so the two profiles are

$$\phi_C(r) = A \frac{\sin(k_0(c_C) \Sigma_{t,C} r)}{\Sigma_{t,C} r}, \quad \phi_R(r) = F \frac{\sinh([b + z_0(c_R) - \Sigma_{t,R} r] / \nu_0(c_R))}{\Sigma_{t,R} r} \quad (3)$$

k_0 and ν_0 are Case's discrete eigenvalue on its two branches, the same roots Assignment 1 used: $c \arctan(k_0) = k_0$ above $c = 1$ and $\text{artanh}(1/\nu_0)/(1/\nu_0) = 1/c$ below it.

Matching, and the one criticality equation

The interface at $r = R_C$ carries a flux relation $\phi_R = g \phi_C$ and continuity of the net current $J = -D d\phi/dr$. In optical variables that current is dimensionless on both sides, $D d\phi/dr = D_0 d/d(\Sigma_t r)$, and the two logarithmic derivatives of ϕ at the interface are $k_0 \cot(k_0 a) - 1/a$ in the core and

$-\nu_0^{-1} \coth([d + z_0]/\nu_0) - 1/(b - d)$ in the reflector. All three approximations therefore reduce to a single transcendental equation for a :

$$D_{0,C} \left[k_0 \cot(k_0 a) - \frac{1}{a} \right] + g D_{0,R} \left[\frac{1}{\nu_0} \coth\left(\frac{d + z_0(c_R)}{\nu_0}\right) + \frac{1}{b - d} \right] = 0 \quad (4)$$

writing $k_0 = k_0(c_C)$ and $\nu_0 = \nu_0(c_R)$. Here $b - d = \Sigma_{t,R} R_C$ is the interface radius counted in *reflector* mean free paths, which is a itself only when the two media share a mean free path; the code keeps the two conversions apart. The reflector enters through nothing but the optical combination $[d + z_0(c_R)]/\nu_0(c_R)$, its thickness in its own diffusion lengths. The two $1/r$ terms have no slab counterpart: ϕ falls as $1/r$ even where u is flat, so a jump in D at a curved interface drives a current by itself. Equation (4) runs from $+\infty$ at $a \rightarrow 0$ to $-\infty$ at $a = \pi/k_0$, the unreflected limit, so the fundamental root is always bracketed by $(0, \pi/k_0)$ and **brentq** finds it in one call.

The three approximations differ only in what they put into D_0 , z_0 and g :

	(a) continuous classic	(b) continuous asymptotic	(c) discontinuous asymptotic
$D_0(c)$	$1/3$	$ 1 - c \nu_0^2$	$ 1 - c \nu_0^2$
relaxation rate	$\sqrt{3 1 - c }$	$k_0(c)$ or $1/\nu_0(c)$	$k_0(c)$ or $1/\nu_0(c)$
$z_0(c)$	$2/3$	$0.710446 [1 + q(1 - c)^2]/c$	$0.710446 [1 + q(1 - c)^2]/c$
$g = \phi_R/\phi_C$	1	1	$\mu_0(c_C)/\mu_0(c_R)$

Table 1: Only the last row makes (c) different from (b). Since $\mu_0 = 1/2$ for every medium in the classic theory, (a) is automatically continuous, and with no reflector at all (b) and (c) are the same bare sphere.

Zimmerman's μ_0 . The discontinuity comes from Zimmerman (UCRL-82792), who writes the partial currents of the asymptotic mode as $\phi_{\pm} = \frac{1}{2}\mu_0\phi_0 \pm \frac{1}{2}\phi_1$, so that $\mu_0\phi_0 = J_+ + J_-$ is the total traffic across a surface and

$$\mu_0(c) = \frac{c\nu_0^2}{2} \ln\left(\frac{\nu_0^2}{\nu_0^2 - 1}\right) \xrightarrow{c>1} \frac{c}{2k_0^2} \ln(1 + k_0^2) \quad (5)$$

the second form being the exact analytic continuation through $\nu_0 = i/k_0$. It is not an ansatz: integrating the discrete mode $\psi(\mu) \propto (\nu_0 - \mu)^{-1}$ over the two half-ranges reproduces it identically, and it returns the two textbook limits $\mu_0 = 1/2$ for an isotropic flux and $\mu_0 = 1$ for a one-way beam. Matching the traffic rather than the flux, $\mu_0(c_C)\phi_C = \mu_0(c_R)\phi_R$, is what makes ϕ itself jump. Nothing is lost at the interface — the same current crosses both faces.

material	c	mfp [cm]	k_0 or $1/\nu_0$	D_0	D [cm]	μ_0	z_0
Pu-239	1.5000	3.064	1.45110	0.23745	0.7275	0.40363	0.47127
U-235	1.3000	3.064	1.05708	0.26847	0.8225	0.43639	0.54552
H ₂ O	0.9000	3.064	0.52543	0.36222	1.1097	0.52660	0.78923
Fe	0.9980	4.300	0.07740	0.33387	1.4356	0.50050	0.71187
Na	1.0000	11.578	0.00000	0.33333	3.8595	0.50000	0.71045

Table 2: Asymptotic parameters of the five materials. Sodium is a pure scatterer, $\Sigma_c = \Sigma_f = 0$, so $c_R = 1$ exactly: ν_0 diverges, its inverse is 0, and the reflector solution degenerates to u_R linear in r , with $\nu_0^{-1} \coth([d + z_0]/\nu_0) \rightarrow [d + z_0]^{-1}$. The code takes that limit explicitly rather than dividing out a near-zero κ .

Results

The bare spheres of Assignment 1 Question 5, which every reflected radius below is measured against:

core	classic R_c [cm]	asympt. R_c [cm]	classic M_c [kg]	asympt. M_c [kg]
Pu-239	5.8163	5.1890	12.940	9.188
U-235	8.1031	7.4339	42.345	32.696

Table 3: Bare reference. Theory (a) is compared with the classical column, (b) and (c) with the asymptotic one.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	5.3159	4.9484	4.5400	9.879	7.968	6.154
Water	2	5.0855	4.7987	4.3823	8.649	7.267	5.535
Water	3	5.0153	4.7499	4.3321	8.296	7.048	5.347
Water	10	4.9814	4.7244	4.3061	8.129	6.935	5.251
Iron	1	5.4212	4.9575	4.6450	10.478	8.013	6.591
Iron	2	5.1056	4.7364	4.4182	8.753	6.988	5.672
Iron	3	4.9506	4.6237	4.3059	7.979	6.501	5.250
Iron	10	4.6790	4.4188	4.1070	6.737	5.674	4.556
Sodium	1	6.4397	5.7159	5.5171	17.562	12.281	11.044
Sodium	2	6.3134	5.6405	5.4318	16.549	11.801	10.540
Sodium	3	6.2505	5.6022	5.3892	16.059	11.563	10.293
Sodium	10	6.1315	5.5286	5.3080	15.159	11.113	9.835

Table 4: Pu-239 core: critical radius in cm and critical mass in kg, under (a) continuous classic, (b) continuous asymptotic and (c) discontinuous asymptotic diffusion. Bare: 5.8163 cm / 12.940 kg classic, 5.1890 cm / 9.188 kg asymptotic.

reflector	d [mfp]	$R_c^{(a)}$	$R_c^{(b)}$	$R_c^{(c)}$	$M_c^{(a)}$	$M_c^{(b)}$	$M_c^{(c)}$
Water	1	7.2501	6.9113	6.5228	30.330	26.273	22.087
Water	2	6.9185	6.6731	6.2694	26.355	23.649	19.611
Water	3	6.8146	6.5937	6.1865	25.186	22.816	18.844
Water	10	6.7638	6.5520	6.1432	24.627	22.385	18.451
Iron	1	7.2811	6.8392	6.5656	30.720	25.460	22.525
Iron	2	6.8002	6.4625	6.1789	25.027	21.480	18.775
Iron	3	6.5545	6.2644	5.9804	22.411	19.565	17.023
Iron	10	6.1125	5.8970	5.6199	18.176	16.320	14.126
Sodium	1	8.4264	7.7951	7.6194	47.618	37.697	35.205
Sodium	2	8.2322	7.6564	7.4692	44.401	35.721	33.164
Sodium	3	8.1328	7.5844	7.3919	42.811	34.722	32.145
Sodium	10	7.9401	7.4425	7.2414	39.840	32.809	30.221

Table 5: U-235 core, same columns. Bare: 8.1031 cm / 42.345 kg classic, 7.4339 cm / 32.696 kg asymptotic.

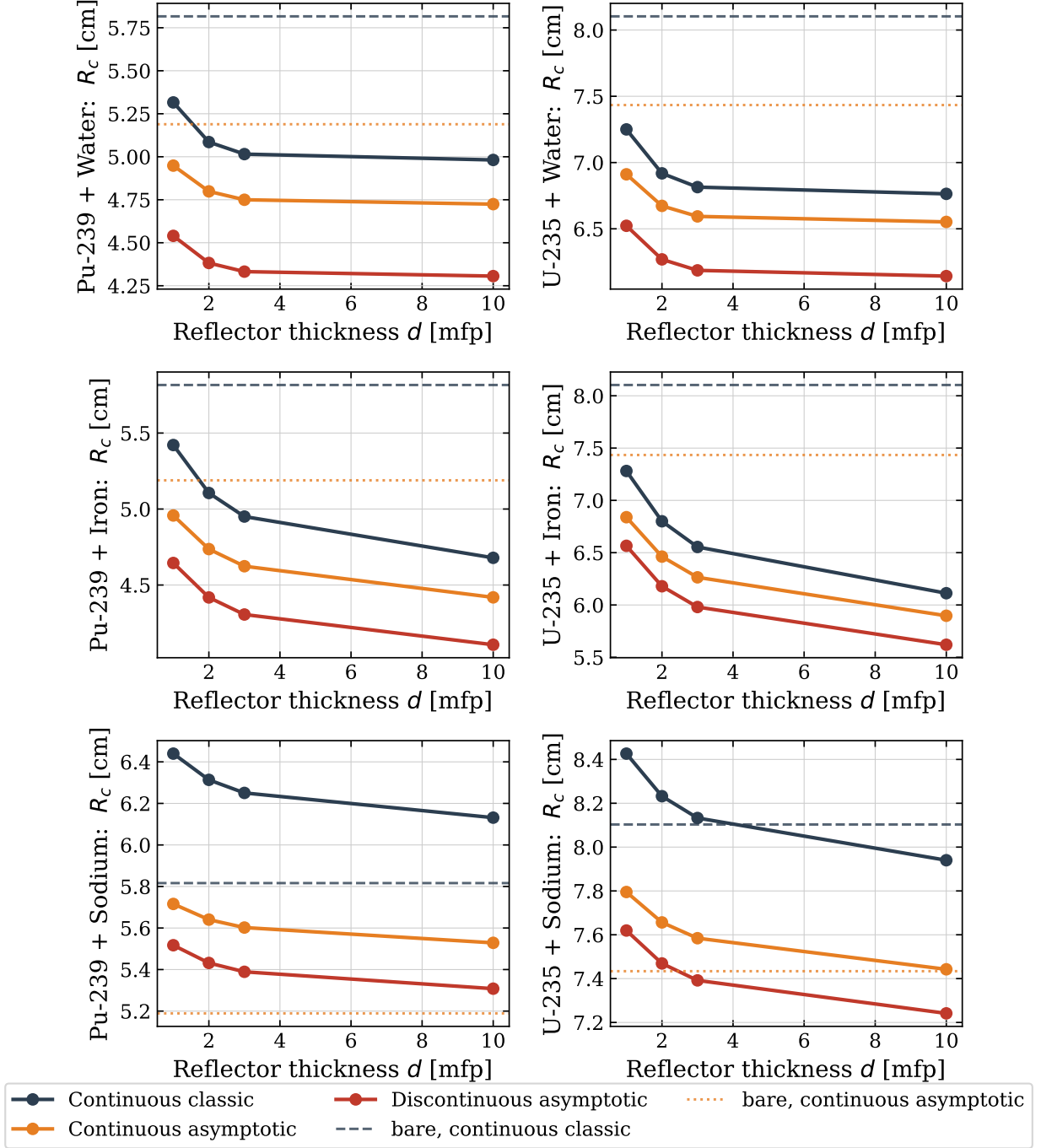


Figure 1: Critical core radius against reflector optical thickness. Dashed and dotted lines are the two bare references. Water saturates by $d \approx 3$ mfp; iron is still improving at $d = 10$; sodium sits above its bare line throughout.

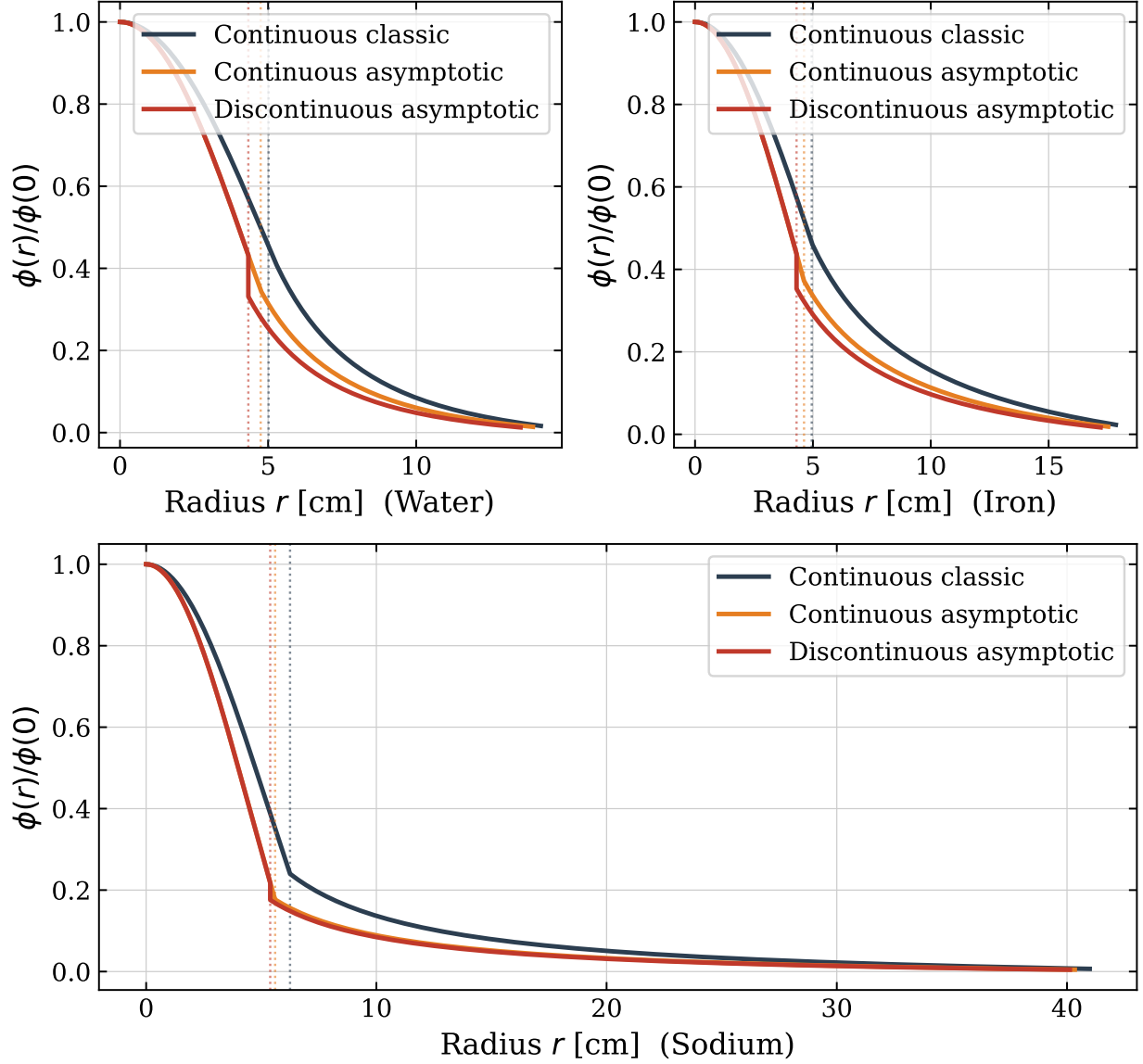


Figure 2: Critical flux across core and reflector for the Pu-239 core behind 3 mfp, normalised to $\phi(0) = 1$; dotted verticals mark each theory's interface. The discontinuous curve drops by the factor g at the interface and is continuous in current, not in flux. Note the horizontal scale of the sodium panel, the wide one beneath: 3 mfp of sodium is 34.7 cm.

What the three approximations say

The reflector saving. Water and iron behave as reflectors should. Both cores lose 12–17% of their radius behind water and 12–24% behind iron by theory (c), which because the mass goes as R^3 is a factor of nearly two in critical mass: Pu-239 falls from 9.188 kg bare to 4.556 kg behind 10 mfp of iron. The two reflectors saturate differently, and the reason is c_R . Water absorbs ($c_R = 0.90$, $\nu_0 = 1.90$), so a neutron more than about two mean free paths deep never returns and the water columns are flat past $d = 3$. Iron barely absorbs ($c_R = 0.998$, $\nu_0 = 12.9$), so its return keeps improving out to $d = 10$ and beyond — iron starts as the worse reflector at $d = 1$ and ends the better one at $d = 10$.

The ordering of the three theories. At every entry $R_c^{(a)} > R_c^{(b)} > R_c^{(c)}$. The first gap is inherited from the bare problem: the asymptotic $D_0(c_C)$ and $k_0(c_C)$ already give a smaller sphere than $D_0 = 1/3$ does, by 11% for Pu-239. The second is the jump itself. Every pair here has $\mu_0(c_C) < \mu_0(c_R)$, so $g < 1$ — 0.766 for Pu-239 behind water, 0.873 for U-235 behind iron — because the multiplying core has the shorter relaxation length and hence the more forward-peaked angular flux, and so carries more current per unit density. A g below one multiplies down both the leakage term and the curvature term of (4), which is why (c) always predicts the smallest core, by a further 8–9% for water and iron.

The sodium columns are diffusion theory failing. Every sodium radius above is *larger* than the corresponding bare sphere, which no reflector can do. The condition is being solved correctly — (4) was checked against an independent two-region spherical finite-volume k -eigenvalue solve, 3000 cells with harmonic-mean face diffusion coefficients across the material jump, and the two agree to better than 0.4% on exactly these cases. The fault is in the model. Splitting the interface leakage of the Pu-239 core at $d = 10$ mfp into its two parts, $J/\phi = D_R/R_c + D_R/L$, the far-face return being $1/L = \Sigma_{t,R} \nu_0^{-1} \coth([d + z_0]/\nu_0)$:

reflector	D_R [cm]	shell [cm]	geometric D_R/R_c	far-face D_R/L	total vs. bare 0.5592
Water	1.021	30.6	0.2050	0.1826	0.3876
Iron	1.433	43.0	0.3063	0.0381	0.3444
Sodium	3.859	115.8	0.6295	0.0312	0.6607

Table 6: Leakage ratio at the interface, in cm^{-1} , classic theory. Sodium’s far-face return is the best of the three — 10 mfp of a pure scatterer really does send almost everything back — but its geometric term alone exceeds what a bare vacuum surface leaks.

The geometric term is the current driven by the $1/r$ spreading of $\phi = u/r$, and it is harmless only while $D_R \ll R_c$. Sodium’s $\Sigma_t = 0.086368 \text{ cm}^{-1}$ is a mean free path of 11.58 cm against a core radius near 6 cm: a neutron entering the reflector travels two core radii before its first collision, which is precisely the regime Fick’s law expands away from. None of the three approximations repairs it, because a pure scatterer has $D_0 = 1/3$ in all of them and $D_R = 3.86 \text{ cm}$ throughout. Measured against its own bare sphere at $d = 10$, (a) is +5.4%, (b) +6.6% and (c) +2.3% — only the Zimmerman jump helps, and only because $g = 0.807$ multiplies the offending term. Transport theory would not make this mistake: the shell is optically thick and purely scattering, so its true albedo is high and the true critical radius must lie below the bare 5.1890 cm.

Question 2: Critical slab half-thickness by P_N

A homogeneous slab of total thickness a , reflective at the midplane $x = 0$ and vacuum at $x = a/2$, in mean free paths ($\Sigma_t = 1$), for $c \in \{1.2, 1.5, 1.8\}$ and $N \in \{1, 3, 5, 9\}$. The unknown is the critical half-thickness $a/2$ at which $k_{\text{eff}} = 1$ — the same quantity Question 3 tabulates for S_N , written the same way, so the two tables compare entry by entry.

The P_N equations

Questions 2–5 all solve the same one-speed slab equation, written out as scattering and fission kept apart, and only the fission source divided by k ,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi(x, \mu) = \frac{1}{2} \left[\Sigma_s \phi(x) + \frac{1}{k} \nu \Sigma_f \phi(x) \right] \quad (6)$$

Since each one of the questions 2–4 specify only c rather than giving Σ_s and $\nu \Sigma_f$ separately, taking the free parameter to be $\Sigma_s = 0$ causes $\nu \Sigma_f = c$ and puts the entire source under k , leaving an equation in c alone:

$$\mu \frac{\partial \psi}{\partial x} + \psi(x, \mu) = \frac{c}{2k} \phi_0(x), \quad \phi_n(x) = \int_{-1}^1 P_n(\mu) \psi(x, \mu) d\mu \quad (7)$$

Explaining why Σ_s can be chosen freely

This question fixes $\Sigma_t = 1$ and $c = (\Sigma_s + \nu \Sigma_f)/\Sigma_t$ and nothing else, so Σ_s is a free parameter: every split $\nu \Sigma_f = c - \Sigma_s$ with $0 \leq \Sigma_s \leq c$ realises the stated c , and they are different media. The split cannot change the quantity asked for, because at $k = 1$ the bracket collapses to $\frac{c}{2} \phi$ whatever it is, so the critical half-thickness is a function of c alone. It does change k at every other half-thickness — Σ_s is not divided by k — so c alone fixes the root of $k(a/2) - 1$ but not $k(a/2)$ itself.

Expanding $\psi(x, \mu) = \sum_{n' \geq 0} \frac{2n'+1}{2} \phi_{n'}(x) P_{n'}(\mu)$, multiplying (7) by $P_n(\mu)$ and integrating over μ turns the streaming term into a three-term coupling through the Legendre recurrence $(2n+1)\mu P_n = (n+1)P_{n+1} + nP_{n-1}$, while the collision term is diagonal and the isotropic source reaches only $n = 0$:

$$\frac{n+1}{2n+1} \frac{d\phi_{n+1}}{dx} + \frac{n}{2n+1} \frac{d\phi_{n-1}}{dx} + \phi_n(x) = \delta_{n0} \frac{c}{k} \phi_0(x), \quad n = 0, \dots, N \quad (8)$$

The system closes on itself once $\phi_{N+1} \equiv 0$ is imposed, which is the entire content of the P_N approximation (also notice that $\tilde{\mathbf{A}}_{\mathbf{0}, -1} = \mathbf{0}$ so this closure over the $N+1$ equations is handed for free). In vector form, with $\Phi = (\phi_0, \dots, \phi_N)^T$,

$$\tilde{\mathbf{A}} \frac{d\Phi}{dx} + \Sigma(k) \Phi = 0, \quad \tilde{A}_{n,n+1} = \frac{n+1}{2n+1}, \quad \tilde{A}_{n,n-1} = \frac{n}{2n+1}, \quad \Sigma_0(k) = \text{diag}(1 - \frac{c}{k}, 1, \dots, 1) \quad (9)$$

$\tilde{\mathbf{A}}$ is the Jacobi matrix of the Legendre recurrence truncated at order N , so its $N+1$ eigenvalues are exactly the roots of $P_{N+1}(\mu)$ — the Gauss–Legendre ordinates of S_{N+1} . That is the algebraic content of the claim in Question 3 that the two methods share their interior equations and differ only at the boundary.

Boundary conditions

At the midplane ($x = 0$) the flux is even in μ , so every odd moment vanishes:

$$\phi_{2m+1}(0) = 0, \quad m = 0, \dots, \frac{N-1}{2} \quad (10)$$

Equivalently $d\phi_{2m}/dx|_0 = 0$, which (8) then delivers for free (Notice that only one of the two families may be imposed or the problem is over-determined.)s

At $x = a/2$ no neutron enters, $\psi(a/2, \mu) = 0$ for $\mu < 0$, which $N + 1$ moments cannot satisfy pointwise. Marshak's choice is to annihilate the $\frac{N+1}{2}$ lowest odd moments of the incoming half-range,

$$\int_{-1}^0 \mu^{2m-1} \psi\left(\frac{a}{2}, \mu\right) d\mu = 0 \implies \sum_{n=0}^N c_{mn} \phi_n\left(\frac{a}{2}\right) = 0, \quad c_{mn} = \frac{2n+1}{2} \int_{-1}^0 \mu^{2m-1} P_n(\mu) d\mu \quad (11)$$

for $m = 1, \dots, \frac{N+1}{2}$. The coefficients follow from $P_n(-\mu) = (-1)^n P_n(\mu)$,

$$c_{mn} = (-1)^{n+1} \frac{2n+1}{2} \int_0^1 \mu^{2m-1} P_n(\mu) d\mu \quad (12)$$

whose integrand is a polynomial of degree at most $2N$, so an $(N + 1)$ -point Gauss–Legendre rule mapped to $[0, 1]$ evaluates it exactly rather than approximately as done in `pn.algebra.marshak_matrix`.

To sum up, (8) is $N + 1$ first-order equations and needs $N + 1$ conditions. (10) supplies $\frac{N+1}{2}$ and (11) the other $\frac{N+1}{2}$.

Box discretisation and the Bell & Glasstone k iteration

The scheme. Divide $[0, a/2]$ into J cells of width $\Delta x = a/(2J)$ with nodes $x_j = j\Delta x$. Integrating (9) over one cell and closing it with the midpoint rule — derivative by the difference across the cell, value by the average of the two faces —

$$\left. \frac{d\phi_n}{dx} \right|_{j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} - \phi_{n,j}}{\Delta x}, \quad \phi_{n,j+\frac{1}{2}} \approx \frac{\phi_{n,j+1} + \phi_{n,j}}{2} \quad (13)$$

gives one $(N + 1)$ -row block per cell,

$$\left(\frac{\tilde{\mathbf{A}}}{\Delta x} + \frac{\mathbf{I}}{2} \right) \Phi_{j+1} + \left(\frac{\mathbf{I}}{2} - \frac{\tilde{\mathbf{A}}}{\Delta x} \right) \Phi_j = q_{j+\frac{1}{2}} \mathbf{e}_0, \quad j = 0, \dots, J-1 \quad (14)$$

The linear system. Stacking the $J + 1$ nodal vectors gives $(J + 1)(N + 1)$ unknowns. Equation (14) supplies $J(N + 1)$ rows, (10) supplies $\frac{N+1}{2}$ rows acting on Φ_0 and (11) the last $\frac{N+1}{2}$ acting on Φ_J . the count closes exactly. Ordered node by node the matrix is block-bidiagonal with two boundary blocks, i.e. banded with half-bandwidth $N + 1$, and one banded (or sparse) LU factorisation solves it. Call that matrix $\mathbf{L}$: it carries streaming, collision and both boundary conditions, and it does *not* depend on k .

The k update. Bell & Glasstone update k by the ratio of successive fission productions,

$$k^{(\ell+1)} = k^{(\ell)} \frac{\int_0^{a/2} \frac{c \Sigma_t}{k^{(\ell)}} \phi_0^{(\ell+1)} dx}{\int_0^{a/2} \frac{c \Sigma_t}{k^{(\ell)}} \phi_0^{(\ell)} dx} = k^{(\ell)} \frac{\int_0^{a/2} \phi_0^{(\ell+1)} dx}{\int_0^{a/2} \phi_0^{(\ell)} dx} \quad (15)$$

The middle form is what the method actually compares: the fission source (14) that $\phi_0^{(\ell+1)}$ would build against the one that produced it, so it is an exact ratio of sources and not of fluxes that happen to resemble them. The constant $c \Sigma_t / k^{(\ell)}$ is common to both and cancels. The iterate is then rescaled by $\max_j \phi_{0,j}$, which has no effect on (15) — it only keeps the vector from drifting towards 0 or ∞ as k^ℓ compounds — and the loop stops on $|k^{(\ell+1)} - k^{(\ell)}| < \epsilon_k$. Convergence is linear at the dominance ratio of the two leading eigenvalues, which for a bare slab is well separated, and the starting guess

$$\phi_0^{(0)}(x) = \cos\left(\frac{\pi x}{2(a/2 + 0.7104)}\right), \quad k^{(0)} = 1 \quad (16)$$

is the diffusion fundamental mode with the asymptotic extrapolation distance, already almost the answer.

The outer loop. Dividing the fission source by k makes a solution exist at every thickness, so criticality is a root search on

$$f(a/2) = k(a/2) - 1 \quad (17)$$

Algorithm 1: Critical half-thickness by the box P_N scheme — homework3.pn.box

```
Input: secondaries  $c$ , order  $N$ , cells  $J$ , initial guess  $(a/2)_0$ 
Output:  $(a/2)_{\text{crit}}$  such that  $k = 1$ 
1 Function BoxSystem( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ):
    /* Setup -- once per  $(a/2, N)$ , outside every loop */
2      $\Delta x \leftarrow (a/2)/J$ ;
3     assemble  $\mathbf{L}$  of order  $(J+1)(N+1)$ :  $\frac{N+1}{2}$  symmetry rows (10) on  $\Phi_0$ ;  $J$  cell blocks (14);
         $\frac{N+1}{2}$  Marshak rows (11) on  $\Phi_J$ ;
4     return splu( $\mathbf{L}$ ) ;           // factorise once; banded, half-bandwidth  $N+1$ 

5 Function k_eigenvalue( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ):
6     system  $\leftarrow$  BoxSystem( $a/2$ ,  $medium$ ,  $N$ ,  $J$ ) ;           // the only factorisation
7      $\phi_0 \leftarrow \cos(\pi x/2(a/2 + 0.7104))$ ,  $k \leftarrow 1$ ;
    /* Power iteration (Bell & Glasstone) */
8     repeat
9          $q \leftarrow \frac{c \Sigma_t}{k}$  midpoints( $\phi_0$ ), in the  $\phi_0$  row of each cell block ;           //  $\phi$  stage
10         $\Phi \leftarrow$  system.solve( $q$ ) ;           // two substitutions, no refactorisation
11         $\phi_0^{\text{new}} \leftarrow$  moment  $n=0$  of  $\Phi$ ;
12         $k_{\text{new}} \leftarrow k \frac{\sum \text{midpoints}(\phi_0^{\text{new}})}{\sum \text{midpoints}(\phi_0)}$  ;           //  $k$  stage, eq. (15)
13         $\phi_0 \leftarrow \phi_0^{\text{new}} / \max_j \phi_{0,j}^{\text{new}}$ ,  $k \leftarrow k_{\text{new}}$  ; // rescale: keeps  $k^\ell$  from compounding
14    until  $|k_{\text{new}} - k| < \epsilon_k |k_{\text{new}}|$ ;
15    return  $k$ 

    /* critical_half_thickness, which defers to sn.core */
16 return critical_size( $a/2 \mapsto k\_eigenvalue(a/2) - 1$ ,  $(a/2)_0$ )
```

Every name in Algorithm 1 is the identifier the code uses. `BoxSystem` is the constructor `pn.box.BoxSystem.__init__`, `splu` is `scipy.sparse.linalg.splu` applied to the CSC form, and `critical_size` is borrowed from `sn.core` — the one place Question 2 reuses the S_N package.

Errors. The box scheme is second order, so the discretisation error in $(a/2)_{\text{crit}}$ falls as Δx^2 and is separable from the truncation error in N by running the same N on a refined mesh — the check reported in Question 3 for S_N .

Equation (9) is a linear system of coupled ODEs with constant coefficients, so it can also be integrated analytically, with no spatial mesh at all. That solution is not needed for the results below and is left to the Appendix, where it serves as the independent check on this one.

Results

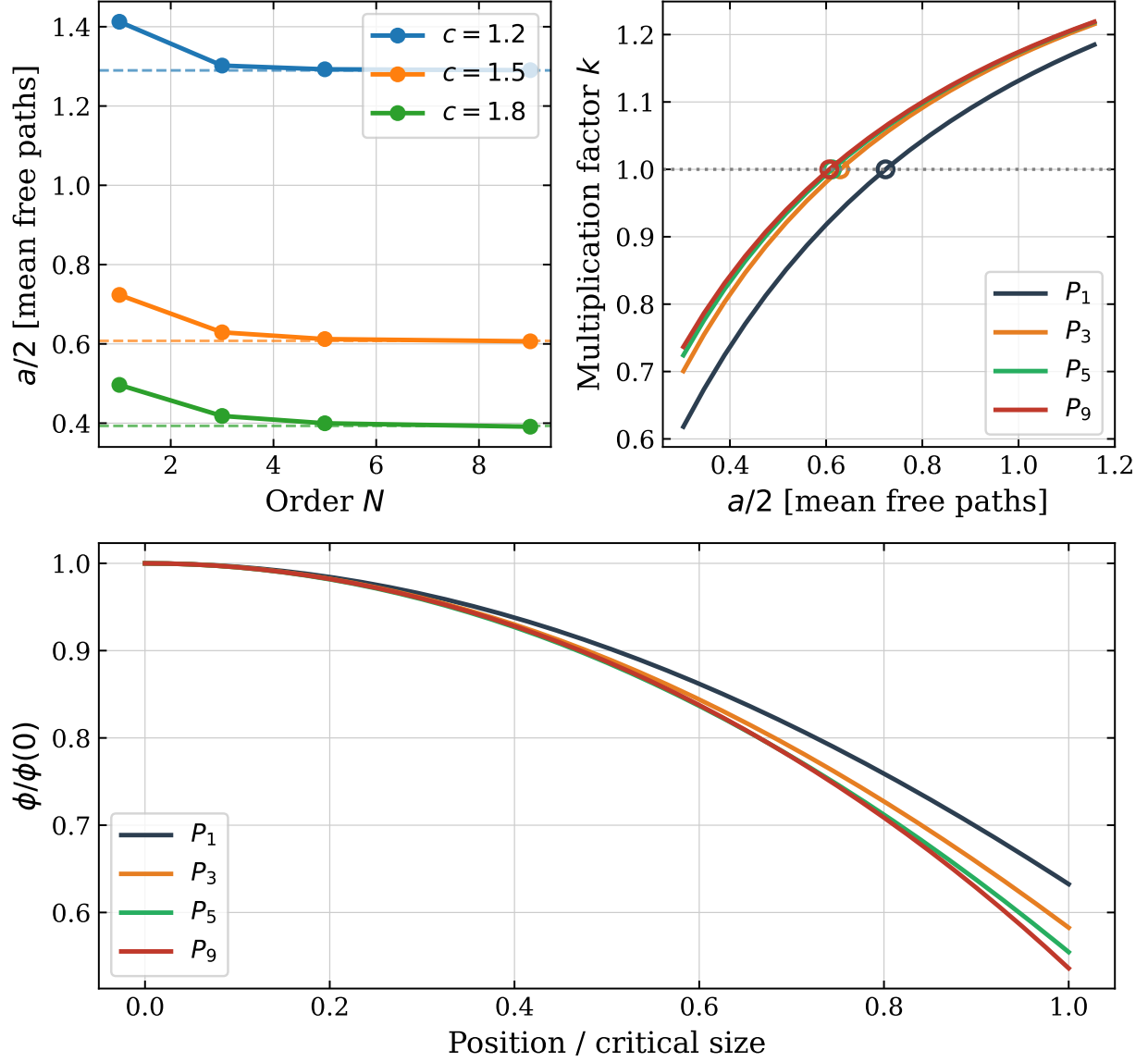


Figure 3: Top left: convergence of $(a/2)_{\text{crit}}$ with N , dashed lines the exact-transport values. Top right: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings, which for P_3 , P_5 and P_9 are already almost on top of one another. Beneath them: the normalised critical flux — as in Question 3, the order changes the surface value and not the interior shape.

c	order	$(a/2)_{\text{crit}}$ [mfp]	exact transport [mfp]	departure
1.2	P_1	1.412502	1.28981	+9.51%
1.2	P_3	1.302001	1.28981	+0.95%
1.2	P_5	1.292597	1.28981	+0.22%
1.2	P_9	1.290382	1.28981	+0.04%
1.5	P_1	0.723480	0.60762	+19.07%
1.5	P_3	0.629149	0.60762	+3.54%
1.5	P_5	0.612136	0.60762	+0.74%
1.5	P_9	0.606467	0.60762	−0.19%
1.8	P_1	0.496560	0.39314	+26.31%
1.8	P_3	0.418210	0.39314	+6.38%
1.8	P_5	0.399736	0.39314	+1.68%
1.8	P_9	0.391130	0.39314	−0.51%

Table 7: Critical half-thickness $(a/2)_{\text{crit}}$ in mean free paths, by the box scheme on 200 cells. The departure is measured against the exact-transport reference $a/2 = \frac{\pi}{2}\nu_0 - z_0$ of Assignment 1.

The S_N method of Questions 3–5

All three questions solve the same monoenergetic, steady-state problem,

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2} \left[\Sigma_s \phi + \frac{1}{k} \nu \Sigma_f \phi \right], \quad \phi(x) = \int_{-1}^1 \psi(x, \mu) d\mu \quad (18)$$

with the same three ingredients: a Gauss–Legendre quadrature in angle, diamond differencing in space with a negative-flux fixup, and the Bell & Glasstone k iteration. Lengths are mean free paths in Questions 3 and 4 and centimetres in Question 5.

Angle. The N ordinates $\{\mu_m\}$ and weights $\{w_m\}$ are the Gauss–Legendre nodes on $[-1, 1]$, so that $\mu_{N-m+1} = -\mu_m$, $\sum_m w_m = 2$, and

$$\phi_i = \sum_{m=1}^N w_m \psi_{i,m} \quad (19)$$

Space, slab. The slab equation is (18) at ordinate m , $\mu_m \partial \psi_m / \partial x + \Sigma_t \psi_m = S/2$, whose conservative form integrates over the cell $x \in [x_{i-1/2}, x_{i+1/2}]$ to

$$\mu_m \left(\psi_{i+\frac{1}{2},m} - \psi_{i-\frac{1}{2},m} \right) + \Sigma_t \Delta x \psi_{i,m} = \frac{S_i}{2} \Delta x \quad (20)$$

with the diamond closure $\psi_{i,m} = \frac{1}{2} (\psi_{i+1/2,m} + \psi_{i-1/2,m})$ relating the cell centre to its two faces. Which face is the outflow depends on the sign of μ_m , and that is the whole of the difference between the two sweeps.

Forward sweep, $\mu_m > 0$: the flux enters at $x_{i-1/2}$ and leaves at $x_{i+1/2}$, so the closure is used as $\psi_{i+1/2,m} = 2\psi_{i,m} - \psi_{i-1/2,m}$. Substituting it into (20) removes the outgoing face,

$$\mu_m \left(2\psi_{i,m} - \psi_{i-\frac{1}{2},m} \right) + \Sigma_t \Delta x \psi_{i,m} = \frac{S_i}{2} \Delta x \quad (21)$$

and collecting $\psi_{i,m}$ gives the cell centre and then, from the closure again, the flux handed to the next cell $i + 1$:

$$\psi_{i,m} = \frac{\frac{1}{2}S_i\Delta x + 2\mu_m\psi_{i-\frac{1}{2},m}}{\Sigma_t\Delta x + 2\mu_m}, \quad \psi_{i+\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i-\frac{1}{2},m} \quad (22)$$

marched from $x = 0$ outwards, starting from the reflected flux there.

Backward sweep, $\mu_m < 0$: the roles swap. Writing $\mu_m = -|\mu_m|$ turns the leading term of (20) into $|\mu_m|(\psi_{i-1/2,m} - \psi_{i+1/2,m})$, so the flux enters at $x_{i+1/2}$ and leaves at $x_{i-1/2}$, and the closure is used the other way round, $\psi_{i-1/2,m} = 2\psi_{i,m} - \psi_{i+1/2,m}$:

$$\psi_{i,m} = \frac{\frac{1}{2}S_i\Delta x + 2|\mu_m|\psi_{i+\frac{1}{2},m}}{\Sigma_t\Delta x + 2|\mu_m|}, \quad \psi_{i-\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i+\frac{1}{2},m} \quad (23)$$

marched from the vacuum face at $x = a/2$ inwards, starting from $\psi_{I+1/2,m} = 0$. The two are the same formula with the inflow face exchanged, which is why one $|\mu_m|$ serves both.

Space, sphere. The spherical equation carries an angular redistribution term, that can be seen in Eq. (2.46) in Pomraning's book (pg. 30):

$$\mu \frac{\partial \psi}{\partial r} + \frac{1 - \mu^2}{r} \frac{\partial \psi}{\partial \mu} + \Sigma_t \psi = \frac{S}{2} \quad (24)$$

whose conservative form integrates over the cell $(r, \mu) \in [r_{i-1/2}, r_{i+1/2}] \times [\mu_{m-1/2}, \mu_{m+1/2}]$ to

$$\mu_m \left(A_{i+\frac{1}{2}} \psi_{i+\frac{1}{2},m} - A_{i-\frac{1}{2}} \psi_{i-\frac{1}{2},m} \right) + \beta_{i,m} \left(\alpha_{m+\frac{1}{2}} \psi_{i,m+\frac{1}{2}} - \alpha_{m-\frac{1}{2}} \psi_{i,m-\frac{1}{2}} \right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \quad (25)$$

with $A = 4\pi r^2$, $V_i = \frac{4\pi}{3}(r_{i+1/2}^3 - r_{i-1/2}^3)$ and the area increment $\beta_{i,m} = (A_{i+1/2} - A_{i-1/2})/w_m$. The angular coefficients follow from demanding that a flat flux solve (25) exactly:

$$\alpha_{\frac{1}{2}} = 0, \quad \alpha_{m+\frac{1}{2}} = \alpha_{m-\frac{1}{2}} - w_m \mu_m, \quad \alpha_{N+\frac{1}{2}} = 0 \quad \text{since} \quad \sum_m w_m \mu_m = 0 \quad (26)$$

The recursion needs a starting value $\psi_{i,1/2}$ at $\mu_{1/2} = -1$, where $1 - \mu^2 = 0$ and the angular derivative drops out; what remains, $-\partial\psi/\partial r + \Sigma_t\psi = S/2$, is (23) at $|\mu| = 1$, run inwards from the vacuum boundary.

The two faces of a spherical cell. Equation (25) is (20) with a second outgoing face. They are not alike and it is worth naming them separately:

- the *radial face*, weighted by the two shell areas $A_{i\pm 1/2}$, whose outflow feeds the **next radial cell** of the same ordinate. Which of the two areas is the outflow depends on the sign of μ_m , exactly as in the slab;
- the *angular face*, weighted by the two coefficients $\alpha_{m\pm 1/2}$ of the angular bin, whose outflow feeds the **next ordinate** in the same cell. This one never swaps: $d\mu/ds \geq 0$ means μ only increases, so the inflow is always at $m - \frac{1}{2}$ and the outflow always at $m + \frac{1}{2}$.

Each closes with its own diamond relation,

$$\psi_{i+\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i-\frac{1}{2},m} \quad (\mu_m > 0), \quad \psi_{i-\frac{1}{2},m} = 2\psi_{i,m} - \psi_{i+\frac{1}{2},m} \quad (\mu_m < 0) \quad (27)$$

$$\psi_{i,m+\frac{1}{2}} = 2\psi_{i,m} - \psi_{i,m-\frac{1}{2}} \quad (28)$$

Take the forward sweep, $\mu_m > 0$, so that the radial outflow is $A_{\text{out}} = A_{i+1/2}$ and the inflow $A_{\text{in}} = A_{i-1/2}$. Substituting (27) and (28) into (25) eliminates both outgoing fluxes at once:

$$\begin{aligned} \mu_m \left(2A_{\text{out}}\psi_{i,m} - (A_{\text{out}} + A_{\text{in}})\psi_{i-\frac{1}{2},m} \right) \\ + \beta_{i,m} \left(2\alpha_{m+\frac{1}{2}}\psi_{i,m} - \left(\alpha_{m+\frac{1}{2}} + \alpha_{m-\frac{1}{2}} \right) \psi_{i,m-\frac{1}{2}} \right) + \Sigma_t V_i \psi_{i,m} = \frac{S_i}{2} V_i \end{aligned} \quad (29)$$

Every term now carries either $\psi_{i,m}$ or a known inflow, so collecting $\psi_{i,m}$ on the left gives the cell solve outright:

$$\boxed{\psi_{i,m} = \frac{\frac{1}{2}S_i V_i + \mu_m (A_{\text{out}} + A_{\text{in}})\psi_{i-\frac{1}{2},m} + \beta_{i,m} \left(\alpha_{m+\frac{1}{2}} + \alpha_{m-\frac{1}{2}} \right) \psi_{i,m-\frac{1}{2}}}{\Sigma_t V_i + 2\mu_m A_{\text{out}} + 2\beta_{i,m} \alpha_{m+\frac{1}{2}}}} \quad (30)$$

and (27) and (28) then hand on the two fluxes the march needs next — $\psi_{i+1/2,m}$ to radial cell $i+1$, and $\psi_{i,m+1/2}$ to ordinate $m+1$. The backward sweep is (30) with $A_{\text{out}} = A_{i-1/2}$, $A_{\text{in}} = A_{i+1/2}$, μ_m replaced by $|\mu_m|$ and the radial inflow taken as $\psi_{i+1/2,m}$; the angular half of the formula is untouched. Setting $A_{\text{out}} = A_{\text{in}} = 1$, $V_i = \Delta x$ and $\alpha_{m\pm 1/2} = 0$ returns (22) exactly, which is the check that the two cell solves are the same balance. **In code they are kept apart on purpose: `sn.slabs.cell_flux` takes one inflow, `sn.sphere.cell_flux` takes a radial face and an angular face.**

Fixup. Whenever a diamond outgoing flux $2\psi_{i,m} - \psi_{\text{in}}$ of (27) or (28) comes out negative it is set to zero and the cell balance is re-solved with it held there, in angle as well as in space. Frankly, this never occurs in Questions 3–5: the fission source is proportional to the flux itself, so no cell is ever starved, and a critical system is thin enough that $\Sigma_t \Delta x / |\mu_m| < 2$ at every ordinate.

Boundary conditions. Reflective at the symmetry point, $\psi_{1/2,m} = \psi_{1/2,m'}$ with $\mu_{m'} = -\mu_m$; vacuum at the outer face, $\psi_{I+1/2,m} = 0$ for $\mu_m < 0$.

Eigenvalue. Outer iteration l fixes the fission source $Q_{f,i}^{(l)} = \nu \Sigma_{f,i} \phi_i^{(l)} / k^{(l)}$; inner iteration n sweeps $S_i = \Sigma_{s,i} \phi_i^{(n)} + Q_{f,i}^{(l)}$ until ϕ settles; then

$$k^{(l+1)} = k^{(l)} \frac{P^{(l+1)}}{P^{(l)}}, \quad P^{(l)} = \sum_i \nu \Sigma_{f,i} \phi_i^{(l)} V_i \quad (31)$$

Dividing the fission source by k makes a solution exist at *every* system size, so criticality becomes a root search: `brentq` on $k(\text{size}) - 1$, which costs 8 evaluations of k .

Algorithm 2: One transport solve — backward ordinates, then forward ones —
`homework3.sn.slab` / `homework3.sn.sphere`

Input: isotropic source S_i

Output: scalar flux ϕ_i

Data: $\psi_{\cdot, m-\frac{1}{2}}$, the *angular inflow column* — one value per spatial cell, the half-angle flux ordinate m consumes through its angular face. Sweeping m replaces it with $\psi_{\cdot, m+\frac{1}{2}}$, which ordinate $m+1$ consumes in turn (`psi_low` and `psi_high` in the code). The slab has no such column.

1 **Function** `sweep_all_angles( $S$ )`:

```

2    $q_i \leftarrow S_i/2$ ;
3   if sphere then  $\psi_{\cdot, \frac{1}{2}} \leftarrow \text{\_starting\_direction}(q)$  //  $\mu_{1/2} = -1$ : no angular term,
   a plane march;
4    $\phi \leftarrow 0$ ;  $\psi_{\frac{1}{2}, \cdot} \leftarrow 0$ ; // the reflective face, one value per ordinate
5   for  $m$  with  $\mu_m < 0$ , ascending do
6   |  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}, \psi_{\frac{1}{2}, m'} \leftarrow \text{sweep\_backward}(m, q, \psi_{\cdot, m-\frac{1}{2}}), \mu_{m'} = -\mu_m$ ;
7   |  $\phi \leftarrow \phi + w_m \psi_{\cdot, m}$ ;
8   for  $m$  with  $\mu_m > 0$ , ascending do
9   |  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}} \leftarrow \text{sweep\_forward}(m, q, \psi_{\cdot, m-\frac{1}{2}}, \psi_{\frac{1}{2}, m})$ ; // starts from what
   |  $-\mu_m$  reflected
10  |  $\phi \leftarrow \phi + w_m \psi_{\cdot, m}$ ;
11  return  $\phi$ 
```

12 **Function** `sweep_backward( $m, q, \psi_{\cdot, m-\frac{1}{2}}$ )`:

```

13   $\psi_{\text{in}} \leftarrow 0$ ; // vacuum: nothing enters the outer face
14  for  $i \leftarrow I$  to 1 (inwards) do
15  | if slab then  $\psi_{i, m}, \psi_{\text{in}} \leftarrow \text{slab.cell\_flux}(\Sigma_t \Delta x, \frac{1}{2} S_i \Delta x, |\mu_m|, \psi_{\text{in}})$  // eq. (23);
16  | else  $\psi_{i, m}, \psi_{\text{in}}, \psi_{i, m+\frac{1}{2}} \leftarrow \text{sphere.cell\_flux}(\Sigma_t V_i, \frac{1}{2} S_i V_i, \text{radial\_face}, \text{angular\_face})$ 
   | // eq. (30);
17  return  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}, \psi_{\text{in}}$ ; // the last  $\psi_{\text{in}}$  is what reflects at the centre
```

18 **Function** `sweep_forward( $m, q, \psi_{\cdot, m-\frac{1}{2}}, \psi_{\text{in}}$ )`:

```

19  for  $i \leftarrow 1$  to  $I$  (outwards) do
20  | the same cell solve, with the radial inflow and outflow faces exchanged; // eq. (22)
21  return  $\psi_{\cdot, m}, \psi_{\cdot, m+\frac{1}{2}}$ 
```

/* The radial face swaps with the sign of μ_m ; the angular face never does
--- its inflow is always $\psi_{i, m-1/2}$ and its outflow always $\psi_{i, m+1/2}$. */

Every name in Algorithms 2 and 3 is the identifier the code uses. `sweep_backward`, `sweep_forward`, `sweep_all_angles` and `\_starting\_direction` are methods of `sn.slabs.SlabSolver` and `sn.spheres.SphereSolver` — the two geometries supply them independently, which is why everything in `sn.core` is written against neither. Three nested loops, and not one matrix among them.

Algorithm 3: The two iterations — scattering inside, Bell & Glasstone outside —
`homework3.sn.core`

Input: medium, geometry, initial guess $size_0$
Output: critical size

```

1 Function run_sn_for_source(solver, medium,  $Q_f$ ,  $\phi$ ):
2   repeat
3      $\phi^{new} \leftarrow \text{sweep\_all\_angles}(\Sigma_s \phi + Q_f)$  ;           // one double sweep per inner
4     until  $\max_i |\phi_i^{new} - \phi_i| \leq \epsilon_\phi \max_i \phi_i^{new}$  or  $\Sigma_s = 0$ ;
5     return  $\phi$  ;           //  $\Sigma_s = 0$  (Q3, Q4): one sweep is already exact

6 Function k_eigenvalue(solver):
7    $\phi \leftarrow 1$ ;  $k \leftarrow 1$ ;  $P \leftarrow \sum_i \nu \Sigma_{f,i} \phi_i V_i$ ;
8   repeat
9      $\phi^{new} \leftarrow \text{run\_sn\_for\_source}(\nu \Sigma_f \phi / k)$  ;           //  $\phi$  iteration
10     $P^{new} \leftarrow \sum_i \nu \Sigma_{f,i} \phi_i^{new} V_i$ ;
11     $k_{new} \leftarrow k P^{new} / P$  ;           //  $k$  iteration
12     $\phi \leftarrow \phi^{new} / \max_i \phi_i^{new}$  ;           // rescale,  $P$  likewise
13  until  $|k_{new} - k| < \epsilon_k |k_{new}|$ ;
14  return  $k$ 

15 Function critical_size(k of size,  $size_0$ ):
16   widen a bracket about  $size_0$  by 3% a step until  $k\_eigenvalue() - 1$  changes sign;
17   return  $\text{brentq}(size \mapsto k\_eigenvalue(size) - 1)$ 

```

Question 3: Critical slab half-thickness by S_N

Reflective at $x = 0$, vacuum at $x = a/2$, 200 cells. The reference column is Assignment 1's exact-transport relation $a/2 = \frac{\pi}{2} \nu_0 - z_0$, evaluated from Case's tabulated ν_0 and z_0 .

The exact one-speed value at $c = 1.5$ is 0.605055 mfp, the Sood *et al.* Pu-239 slab (1.853722 cm at $\Sigma_t = 0.32640 \text{ cm}^{-1}$, the same table Assignment 1 Question 5 uses). The S_N sequence walks down onto it monotonically: 0.609042 at S_{10} , 0.606407 at S_{16} , 0.605631 at S_{24} , 0.605195 at S_{48} .

P_N against S_{N+1}

In slab geometry the two are algebraically the same interior equations. Expanding ψ in $N + 1$ Legendre moments and truncating gives the same $N + 1$ coupled ODEs as collocating at the $N + 1$ Gauss–Legendre points, because a Gauss–Legendre rule of order $N + 1$ integrates polynomials up to degree $2N + 1$ exactly. **Any difference between P_N and S_{N+1} is therefore a difference of boundary condition**, and it is largest at low N where the boundary is the whole story:

- S_{N+1} sets $\psi(a/2, \mu_m) = 0$ on each incoming ordinate — the *Mark* condition.
- P_N needs conditions on moments instead. *Marshak*'s, $\int_0^1 P_{2j+1}(\mu) \psi d\mu = 0$, weight the whole incoming half-range; *Mark*'s pin the same isolated directions S_{N+1} does.

With *Mark* conditions the two agree; with *Marshak* they do not. Concretely at $c = 1.5$: S_2 above gives $a/2 = 0.78001$, and P_1 with the *Mark* extrapolation $l_0 = 1/\sqrt{3}$ gives $\arctan(1/Bl_0)/B = 0.78001$ — the same five digits — while P_1 with the *Marshak* $l_0 = 2/3$ gives 0.72348, a 7% spread between two methods that share every interior equation.

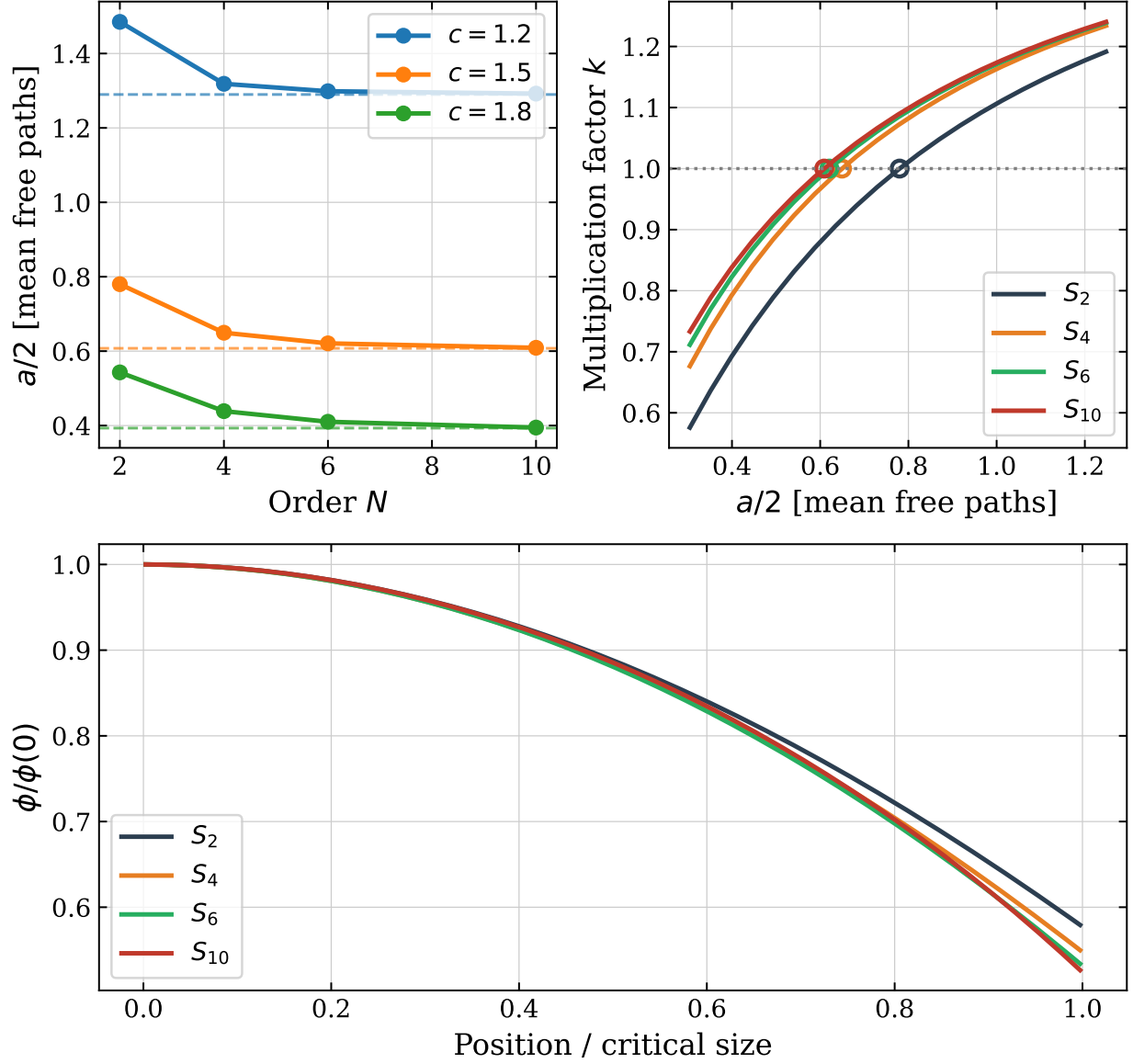


Figure 4: Top left: convergence of $a/2$ with N , dashed lines the exact-transport values. Top right: k against half-thickness at $c = 1.5$, circles the $k = 1$ crossings. Beneath them: the normalised critical flux, which barely moves with N — the order affects the boundary, not the interior.

c	order	$a/2$ [mfp]	exact transport [mfp]	departure
1.2	S_2	1.48499	1.28981	+15.13%
1.2	S_4	1.31855	1.28981	+2.23%
1.2	S_6	1.29861	1.28981	+0.68%
1.2	S_{10}	1.29218	1.28981	+0.18%
1.5	S_2	0.78001	0.60762	+28.37%
1.5	S_4	0.64949	0.60762	+6.89%
1.5	S_6	0.62089	0.60762	+2.18%
1.5	S_{10}	0.60904	0.60762	+0.23%
1.8	S_2	0.54291	0.39314	+38.09%
1.8	S_4	0.43875	0.39314	+11.60%
1.8	S_6	0.41014	0.39314	+4.32%
1.8	S_{10}	0.39459	0.39314	+0.37%

Table 8: Critical half-thickness in mean free paths. The departure is entirely angular: the same radii are reproduced to seven digits on meshes from 50 to 800 cells.

Question 2 supplies the rest of that comparison, since it solves the same slab at $N = 1, 3, 5, 9$ under Marshak conditions:

c	pair	P_N (Marshak)	S_{N+1} (Mark)	$S_{N+1}/P_N - 1$	exact transport
1.2	P_1 / S_2	1.41250	1.48499	+5.13%	1.28981
1.2	P_3 / S_4	1.30200	1.31855	+1.27%	1.28981
1.2	P_5 / S_6	1.29259	1.29861	+0.47%	1.28981
1.2	P_9 / S_{10}	1.29038	1.29218	+0.14%	1.28981
1.5	P_1 / S_2	0.72348	0.78001	+7.81%	0.60762
1.5	P_3 / S_4	0.62915	0.64949	+3.23%	0.60762
1.5	P_5 / S_6	0.61213	0.62089	+1.43%	0.60762
1.5	P_9 / S_{10}	0.60647	0.60904	+0.42%	0.60762
1.8	P_1 / S_2	0.49656	0.54291	+9.33%	0.39314
1.8	P_3 / S_4	0.41821	0.43875	+4.91%	0.39314
1.8	P_5 / S_6	0.39974	0.41014	+2.60%	0.39314
1.8	P_9 / S_{10}	0.39113	0.39459	+0.88%	0.39314

Table 9: The two methods at matched order, in mean free paths. Every P_N entry is below its S_{N+1} partner and closer to the exact value, and the gap closes as N grows.

Three things in that table follow from the argument above. **The sign of the gap is the sign of l_0 :** Marshak’s extrapolation $2/3$ is longer than Mark’s $1/\sqrt{3} = 0.577$, so the P_N flux is driven to zero further outside the slab and less material is needed — P_N is below S_{N+1} at every one of the twelve entries. **The gap is a boundary effect, so it shrinks with N ,** from 5–9% at $N = 1$ to under 1% at $N = 9$; by then both boundary conditions are resolving the same half-range flux and the choice between them stops mattering. **And it grows with c at fixed N** — 5.13%, 7.81%, 9.33% at $N = 1$ — because a larger c means a thinner slab, a shorter relaxation length and a more forward-peaked surface flux, which is exactly the shape a half-range condition of low order cannot

capture. The whole difference is worth at most a factor of two in either method's error: at $c = 1.5$, P_9 is +0.23% from the true 0.605055 mfp where S_{10} is +0.66%, so a P_N solve buys roughly the accuracy of an S_N solve of the next order up.

All of that is a statement about *slabs*. In the sphere the equivalence fails at the interior equations themselves: the angular derivative $\frac{1-\mu^2}{r} \frac{\partial \psi}{\partial \mu}$ couples the Legendre moments in a different pattern from the way the α recursion of (25) couples the ordinates, so P_1 and S_2 are genuinely different approximations there rather than the same one under two boundary conditions.

Question 4: Critical sphere radius by S_N

Reflective at $r = 0$, vacuum at $r = R$, 100 cells, with the angular redistribution term of (25). The reference is Assignment 1's $\Sigma_t R_c = \pi \nu_0 - z_0$.

c	order	R_c [mfp]	exact transport [mfp]	departure
1.2	S_2	3.06060	3.17204	−3.51%
1.2	S_4	3.14631	3.17204	−0.81%
1.2	S_6	3.15991	3.17204	−0.38%
1.2	S_{10}	3.16728	3.17204	−0.15%
1.5	S_2	1.60725	1.69010	−4.90%
1.5	S_4	1.66739	1.69010	−1.34%
1.5	S_6	1.67934	1.69010	−0.64%
1.5	S_{10}	1.68595	1.69010	−0.25%
1.8	S_2	1.11753	1.18295	−5.53%
1.8	S_4	1.16413	1.18295	−1.59%
1.8	S_6	1.17412	1.18295	−0.75%
1.8	S_{10}	1.17971	1.18295	−0.27%

Table 10: Critical radius in mean free paths. Note the sign: the sphere converges from *below*, the slab from above.

Cross-check between the two geometries. Both drive the asymptotic flux to zero at the same extrapolated surface, so $a/2 + z_0 = \frac{\pi}{2} \nu_0$ and $R_c + z_0 = \pi \nu_0$, and therefore

$$z_0 = R_c - 2(a/2) \tag{32}$$

which lets the two tables above check each other without any further calculation:

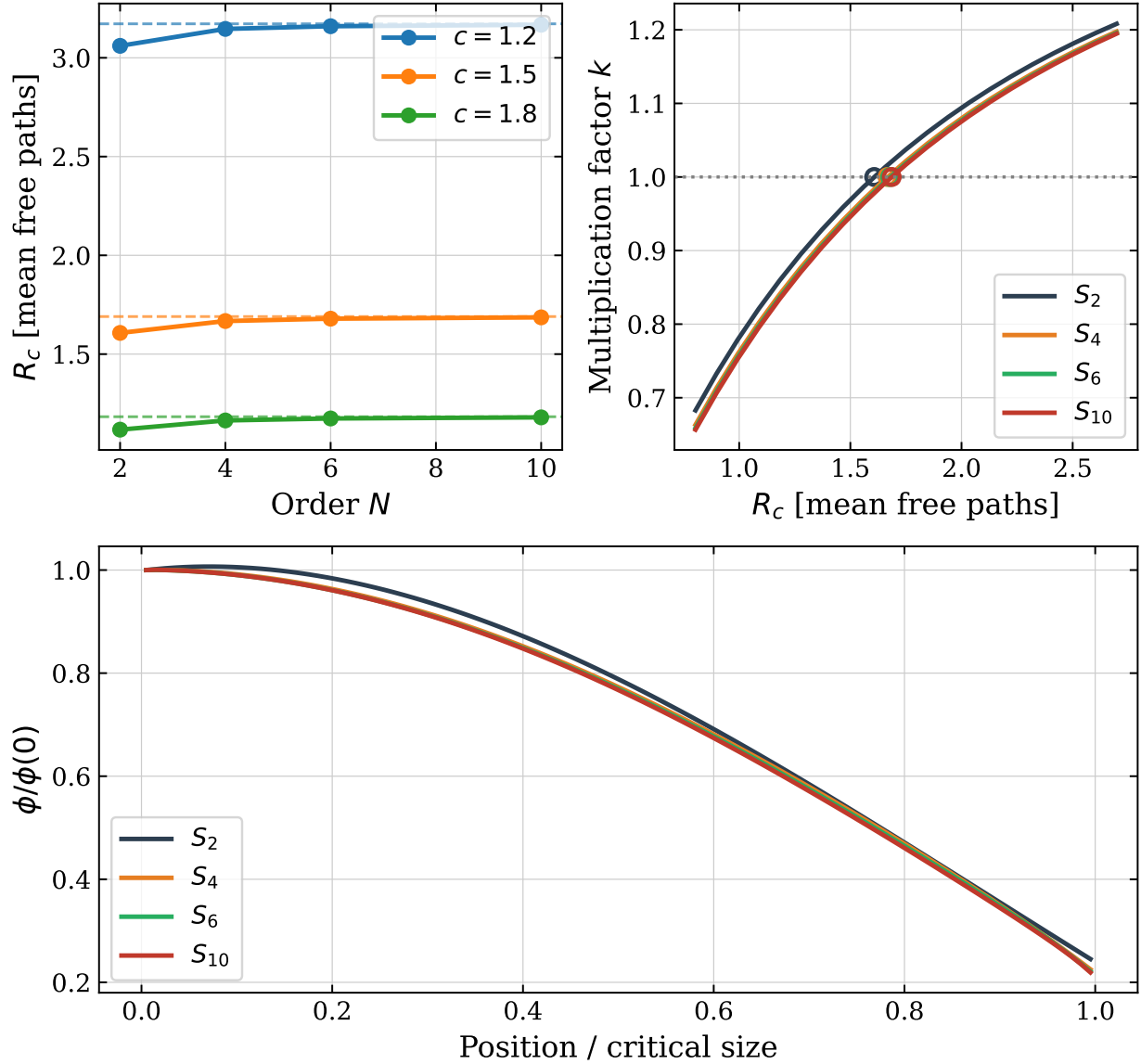


Figure 5: The same three panels for the sphere. The S_2 curve is the outlier in all three: in curvilinear geometry S_2 is not the P_1 equation, and the angular redistribution term it misrepresents is a loss from the interior directions.

c	from S_2	from S_4	from S_6	from S_{10}	Case's table
1.2	0.091	0.509	0.563	0.583	0.592
1.5	0.047	0.368	0.438	0.468	0.475
1.8	0.032	0.287	0.354	0.391	0.397

Table 11: Extrapolation distance z_0 implied by the two geometries, in mean free paths. At S_{10} it is 1.5% below Case's value at all three c — the same shortfall three times, which is the relation's own asymptotic error. At S_2 it nearly vanishes, which is the sharpest statement of what low-order S_N gets wrong: the boundary, not the interior.

Question 5: Critical mass of U-235 and Pu-239 by S_N

The cross sections are the Question 5 table of Assignment 1 — $\Sigma_t = 0.32640 \text{ cm}^{-1}$ for both, giving $c = 1.500$ for Pu-239 and $c = 1.300$ for U-235 — and the critical mass follows from the converged radius as $M_c = \frac{4}{3}\pi R_c^3 \rho$. Here $\Sigma_s > 0$, so the inner iteration does real work: about ten sweeps per outer, against one for the c -only problems above. The orders are S_2 to S_{16} rather than the S_2 – S_{10} of the previous questions, since the mass goes as the cube of the radius and is the more demanding quantity.

material	order	c	R_c [cm]	M_c [kg]	asympt. diffusion R_c [cm]	departure
Pu-239	S_2	1.500	4.9242	7.852	5.1890	−5.10%
Pu-239	S_4	1.500	5.1084	8.767	5.1890	−1.55%
Pu-239	S_8	1.500	5.1586	9.028	5.1890	−0.59%
Pu-239	S_{16}	1.500	5.1730	9.104	5.1890	−0.31%
U-235	S_2	1.300	7.1218	28.748	7.4339	−4.20%
U-235	S_4	1.300	7.3514	31.619	7.4339	−1.11%
U-235	S_8	1.300	7.4072	32.345	7.4339	−0.36%
U-235	S_{16}	1.300	7.4231	32.553	7.4339	−0.15%

Table 12: Critical radius and mass, against the asymptotic diffusion result of Assignment 1 Question 5. The mass is the cube of the radius, so the 5% radius error of S_2 is a 14% error in mass.

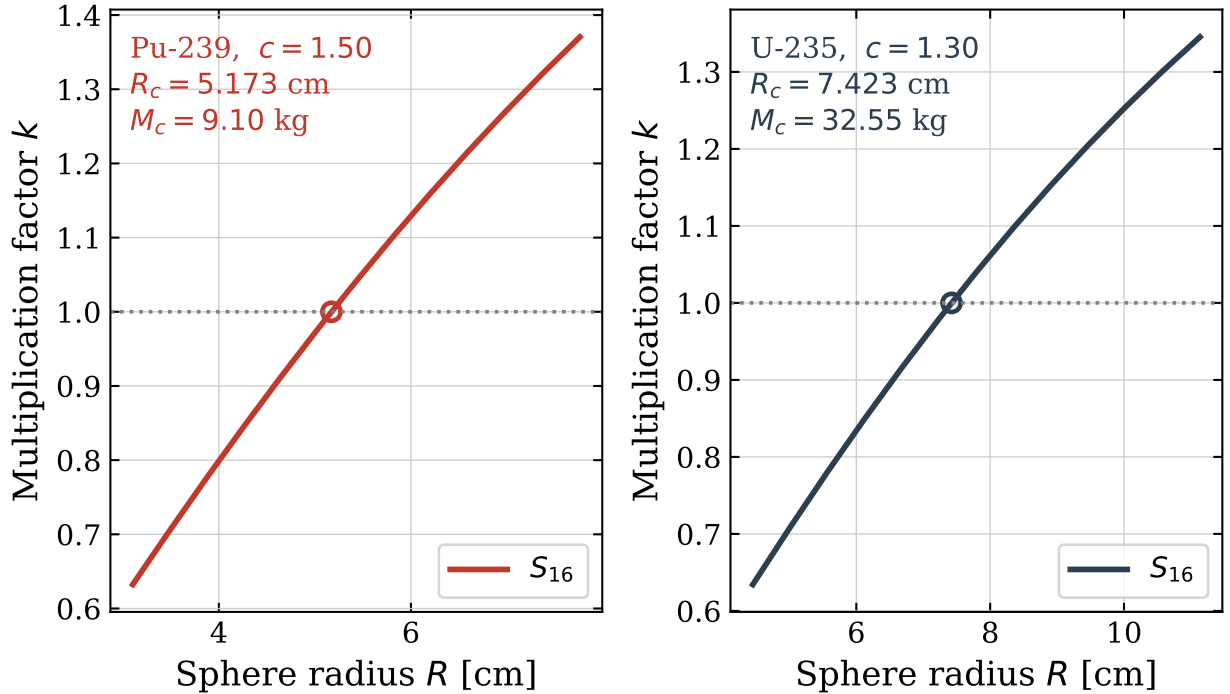


Figure 6: k against sphere radius at S_{16} , with the $k = 1$ crossing that fixes the critical mass.

Diffusion overestimates both radii, and by more for Pu-239 than for U-235: the surface sits about two mean free paths from the centre in both cases, but the higher c makes the Pu-239 flux

fall off faster and its surface angular flux more forward-peaked, which is exactly what a diffusion approximation cannot represent. S_{16} recovers the difference to within 0.3% of the transport-theory radius.

Appendix: the coupled P_N ODE system solved analytically

The box scheme of Question 2 carries two errors, one in N and one in Δx . Since (9) is a linear system of coupled ODEs with constant coefficients, it can be integrated in closed form at $k = 1$ instead, with criticality imposed directly on the boundary conditions. That removes the Δx error altogether and leaves only the truncation in N , which is what makes it an independent check on the mesh solution rather than a repetition of it.

Split the $N + 1$ moments into $M = \frac{N+1}{2}$ even and M odd, $\Phi_{\text{even}} = (\phi_0, \phi_2, \dots)$ and $\Phi_{\text{odd}} = (\phi_1, \phi_3, \dots)$. Every derivative in (8) changes the parity of its index, so the even- n and odd- n equations separate:

$$\mathbf{A} \frac{d\Phi_{\text{odd}}}{dx} + \Sigma_0(k) \Phi_{\text{even}} = 0, \quad \mathbf{B} \frac{d\Phi_{\text{even}}}{dx} + \Phi_{\text{odd}} = 0, \quad \Sigma_0(k) = \text{diag}\left(1 - \frac{c}{k}, 1, \dots, 1\right) \quad (33)$$

$\mathbf{A}$ and $\mathbf{B}$ being the two off-diagonal blocks of $\tilde{\mathbf{A}}$ under that reordering, $\tilde{\mathbf{A}} = \begin{pmatrix} \mathbf{0} & \mathbf{A} \\ \mathbf{B} & \mathbf{0} \end{pmatrix}$, with rows $A_{m,\cdot}$ from $n = 2m$ and $B_{m,\cdot}$ from $n = 2m + 1$. Substituting $\Phi_{\text{odd}} = -\mathbf{B} \Phi'_{\text{even}}$ into the first equation eliminates the odd moments and leaves a coupled Helmholtz system,

$$\frac{d^2 \Phi_{\text{even}}}{dx^2} + \mathbf{K}^2(k) \Phi_{\text{even}} = 0, \quad \mathbf{K}^2(k) = -(\mathbf{AB})^{-1} \Sigma_0(k) \quad (34)$$

The minus sign is not cosmetic and is the easiest thing to lose: the elimination produces $\mathbf{AB} \Phi''_{\text{even}} = \Sigma_0 \Phi_{\text{even}}$, and moving that to the form (34) flips it. At $N = 1$ it reads $\phi''_0 + 3(c - 1)\phi_0 = 0$, the classic diffusion buckling $B = \sqrt{3(c - 1)}$ of the table in Question 1 — with the sign dropped one gets cosh where cos belongs and no critical thickness at all.

The spectrum of $\mathbf{K}^2$. Since $\tilde{\mathbf{A}}^2 = \begin{pmatrix} \mathbf{AB} & \mathbf{0} \\ \mathbf{0} & \mathbf{BA} \end{pmatrix}$ and the eigenvalues of $\tilde{\mathbf{A}}$ are the $N + 1$ roots $\pm \mu_i$ of P_{N+1} , the matrix $\mathbf{AB}$ has the M eigenvalues $\mu_i^2 > 0$ and is similar to a positive definite matrix. By Sylvester's law of inertia the eigenvalues of (34) then carry the signs of $-\Sigma_0$, which for $c > 1$ at $k = 1$ has one negative and $M - 1$ positive entries: **exactly one eigenvalue** $\lambda_1^2 = B_0^2 > 0$, **and** $M - 1$ **eigenvalues** $\lambda_j^2 = -\kappa_j^2 < 0$. The one oscillatory mode is the fundamental; the rest are the boundary layers, and at $c = 1$ the positive one collapses to $B_0 = 0$, an infinite critical slab.

Assembling the criticality condition. Diagonalise $\mathbf{K}^2(1) = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^{-1}$. The solution even about $x = 0$, which satisfies (10) identically because $\Phi_{\text{odd}} = -\mathbf{B} \Phi'_{\text{even}}$ vanishes wherever Φ'_{even} does, is

$$\Phi_{\text{even}}(x) = C_1 \mathbf{v}_1 \cos(B_0 x) + \sum_{j=2}^M C_j \mathbf{v}_j \cosh(\kappa_j x) \quad (35)$$

$$\Phi_{\text{odd}}(x) = C_1 (\mathbf{B} \mathbf{v}_1) B_0 \sin(B_0 x) - \sum_{j=2}^M C_j (\mathbf{B} \mathbf{v}_j) \kappa_j \sinh(\kappa_j x) \quad (36)$$

M free constants against the M Marshak conditions (11). Splitting the coefficient matrix c_{mn} into its even and odd columns $\mathbf{M}_{\text{even}}, \mathbf{M}_{\text{odd}}$ and evaluating at $x = a/2$ gives M homogeneous equations

$\mathbf{H}(a/2)\mathbf{C} = \mathbf{0}$ with

$$\begin{aligned}\mathbf{H}_{.1} &= \mathbf{M}_{\text{even}} \mathbf{v}_1 \cos\left(\frac{B_0 a}{2}\right) + \mathbf{M}_{\text{odd}}(\mathbf{B} \mathbf{v}_1) B_0 \sin\left(\frac{B_0 a}{2}\right) \\ \mathbf{H}_{.j} &= \mathbf{M}_{\text{even}} \mathbf{v}_j \cosh\left(\frac{\kappa_j a}{2}\right) - \mathbf{M}_{\text{odd}}(\mathbf{B} \mathbf{v}_j) \kappa_j \sinh\left(\frac{\kappa_j a}{2}\right), \quad j \geq 2\end{aligned}$$

and a non-trivial flux exists only where

$$\boxed{\det \mathbf{H}(a/2) = 0} \tag{37}$$

$(a/2)_{\text{crit}}$ is its smallest positive root. Two numerical points: the cosh and sinh columns overflow once $\frac{1}{2}\kappa_j a \gtrsim 700$, so each is divided by its own $\cosh(\frac{1}{2}\kappa_j a)$, which leaves $\mathbf{v}_j$ and $-(\mathbf{B} \mathbf{v}_j) \kappa_j \tanh(\frac{1}{2}\kappa_j a)$ and cannot move a zero of the determinant, a column scaling being a non-zero factor of it; and the determinant itself spans many orders of magnitude across the bracket, so it is the sign change that is tracked, not the value.

Against the box scheme. The two solutions are the only free check on each other, since they share nothing but the matrices of (9) and (11): one iterates a discretised operator, the other never discretises space at all. They agree to between one and three parts in 10^6 , and that residue is the box scheme's Δx^2 and nothing else. Refining the mesh at $c = 1.5$, $N = 5$ against the analytic 0.612134509:

cells J	25	50	100	200	400
error [mfp]	$6.61 \cdot 10^{-5}$	$1.65 \cdot 10^{-5}$	$4.13 \cdot 10^{-6}$	$1.03 \cdot 10^{-6}$	$2.58 \cdot 10^{-7}$
ratio	—	4.00	4.00	4.00	4.00

Each halving of Δx divides the error by 4.00 to three digits, which is the second order claimed in Question 2 measured rather than asserted, and it confirms that the boundary rows are as accurate as the interior ones — a first-order boundary closure would have shown up here as a ratio of 2.